EXP-2

EXPT NO: 2 OCC AND LOAD CHARACTERISTICS OF D.C SHUNT GENERATOR

Aim:

To conduct load test on separately excited generators and to obtain the characteristics.

Apparatus Required:

The apparatus required for the experiment are one voltmeter with a range of 0–300 V (MC type), one ammeter with a range of 0–2.5 A (MC type), one ammeter with a range of 0–5 A (MC type), two wire-wound rheostats, and connecting wires as required.

Fuse Rating Calculation for Field and Armature:

No-load test: 10% of rated current (full-load current).

Load test: 125% of rated current (full-load current).

Precautions:

1. Motor side field rheostat should be kept at minimum resistance position.
2. Generator side field rheostat should be kept at maximum resistance position.
3. Starter should be in off position before switching on the supply.
4. The DPST switch must be kept open.

Procedure for Open Circuit test:

1. Connections are given as per the circuit diagram.
2. The motor is started with the help of THREE POINT starter.
3. Adjust the motor speed to rated speed by adjusting motor field rheostat when the generator is disconnected from the load by DPST switch 2.
4. By varying the generator field rheostat gradually, the open circuit voltage [Eo] and corresponding field current (If) are tabulated up to 125 % of rated voltage of generator.
5. The motor is switched off by using DPST switch 1 after bringing all the rheostats to initial position.

Procedure for Load test:

1. Connections are given as per the circuit diagram
2. The prime mover is started with the help of three point starter and it is made to run at rated speed when the generator is disconnected from the load by DPST switch 2.
3. By varying the generator field rheostat gradually, the rated voltage [Eg] is obtained.
4. The ammeter and voltmeter readings are observed at no load condition.
5. The ammeter and voltmeter readings are observed for different loads up to the rated current by closing the DPST switch 2.
6. After tabulating all the readings the load is brought to its initial position.
7. The motor is switched off by using DPST switch 1 after bringing all the rheostats to initial position.

Tabulation for Open Circuit Test:

S.NO	Field Current I_f (A)	Open Circuit Voltage E_0 (V)
1.	0	0
2.	0.16	115
3.	0.18	120
4.	0.20	126
5.	0.23	133
6.	0.24	135

Calculation:

CALCULATION FOR OPEN CIRCUIT CHARACTERISTICS:

formula for voltage-to-current ratio :

$$R = \frac{E_0}{I_f}$$

Reading from 1-6 :

$R_1 = 0$

$R_2 = \frac{115}{0.16} = 718.75$

$R_3 = \frac{120}{0.18} = 666.67$

$R_4 = \frac{126}{0.20} = 630$

$R_5 = \frac{133}{0.23} = 578.26$

$R_6 = \frac{135}{0.24} = 562.5$

Tabulation for Load Test:

S.NO	Load Current (A)	Terminal Voltage (V)
1.	1.2	220
2.	2.8	212
3.	3.6	205
4.	7.0	195
5.	10.2	184
6.	12.7	176

Calculation:

CALCULATION FOR LOAD CHARACTERISTICS:

Formula for output power:

$$P_{out} = VI_L$$

Reading from 1 – 6 :

$P_1 = 220 \times 1.2 = 264\ W$

$P_2 = 212 \times 2.8 = 593.6\ W$

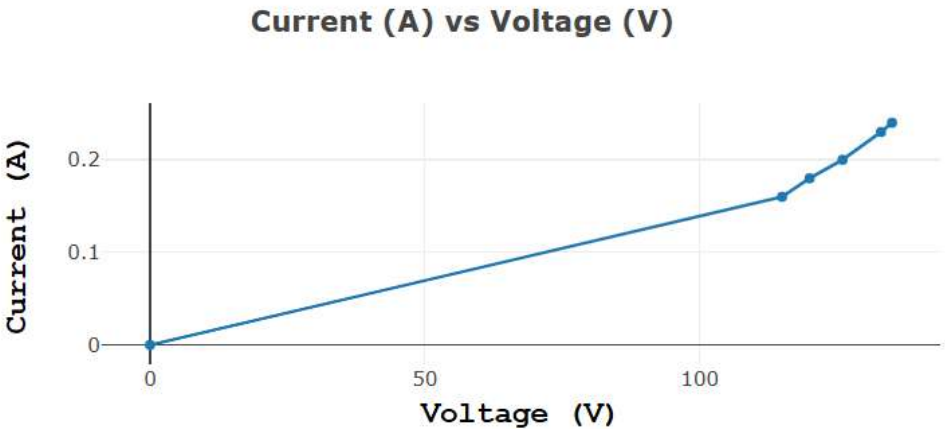
$P_3 = 205 \times 3.6 = 738\ W$

$P_4 = 195 \times 7.0 = 1365\ W$

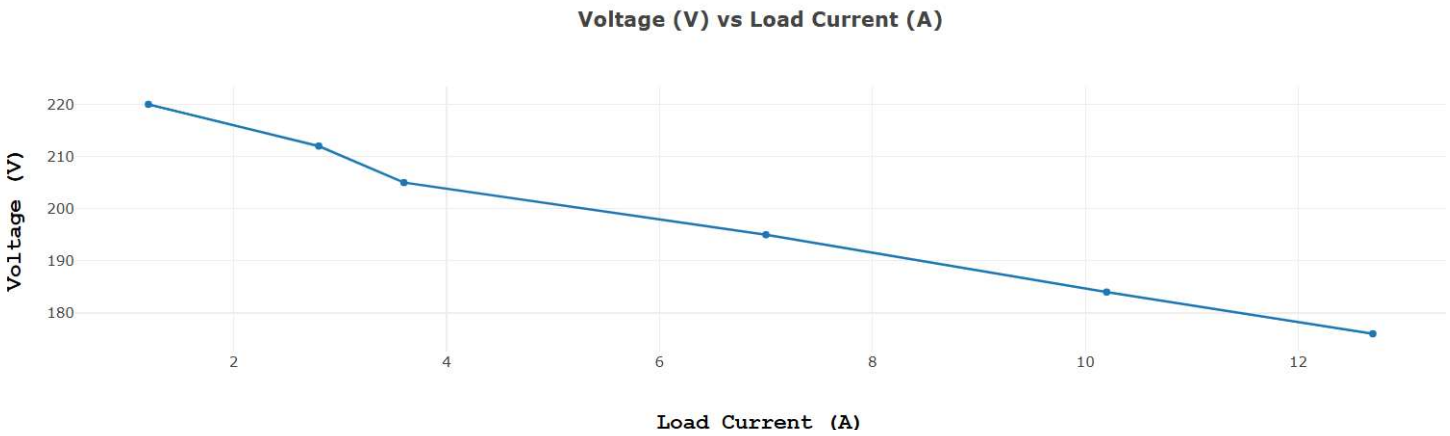
$P_5 = 184 \times 10.2 = 1876.8\ W$

$P_6 = 176 \times 12.7 = 2235.2\ W$

Open Circuit Characteristics:



Load Characteristics:



Result:

The load test on separately excited generators and to obtain the characteristics was verified.